Here we study some common special series like

(i) Sum of first 'n' natural numbers

(ii) Sum of first 'n' odd natural numbers.

(iii) Sum of squares of first 'n' natural numbers.

(iv) Sum of cubes of first 'n' natural numbers.

We can derive the formula for sum of any powers of first n natural numbers using the expression $(x+1)^{k+1} - x^{k+1}$. That is to find $1^k + 2^k + 3^k + ... + n^k$ we can use the expression $(x+1)^{k+1} - x^{k+1}$.

2.11.1 Sum of first *n* natural numbers

To find $1+2+3+\cdots+n$, let us consider the identity $(x+1)^2 - x^2 = 2x+1$

Where x = 1, 2, 3,.....n – 1, n

$$x=1,\ 2^2-1^2=2(1)+1$$
$$x=2,\ 3^2-2^2=2(2)+1$$
$$x=3,\ 4^2-3^2=2(3)+1$$
$$\vdots \qquad \vdots \qquad \vdots$$
$$x=n-1,\ n^2-(n-1)^2=2(n-1)+1$$
$$x=n,\ (n+1)^2-n^2=2(n)+1$$

Adding all these equations and cancelling the terms on the Left Hand side, we get,

$$\begin{aligned}(n+1)^2-1^2 &= 2(1+2+3+\cdots+n)+n\\ n^2+2n &= 2(1+2+3+\cdots+n)+n\\ 2(1+2+3+\cdots+n) &= n^2+n=n(n+1)\\ 1+2+3+\cdots+n &= \frac{n(n+1)}{2}\end{aligned}$$

2.11.2 Sum of first *n* odd natural numbers

$1+3+5+\cdots+(2n-1)$

It is an A.P. with $a=1$, $d=2$ and $l=2n-1$

$$\begin{aligned}S_n &= \frac{n}{2}[a+l]\\ &= \frac{n}{2}[1+2n-1]\\ S_n &= \frac{n}{2}\times 2n = n^2\end{aligned}$$

2.11.3 Sum of squares of first *n* natural numbers

To find $1^2+2^2+3^2+\cdots+n^2$, let us consider the identity $(x+1)^3 - x^3 = 3x^2+3x+1$

Where x = 1, 2, 3,.....n – 1, n

$$x=1,\ 2^3-1^3=3(1)^2+3(1)+1$$
$$x=2,\ 3^3-2^3=3(2)^2+3(2)+1$$
$$x=3,\ 4^3-3^3=3(3)^2+3(3)+1$$
$$\vdots \qquad \vdots \qquad \vdots$$
$$x=n-1,\ n^3-(n-1)^3=3(n-1)^2+3(n-1)+1$$
$$x=n,\ (n+1)^3-n^3=3n^2+3n+1$$

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